

Evaluating Prompt Engineering Techniques for Automated BDD Test Generation from Connextra User Stories

Group 2 - SE1905 - SWT301

- Ngô Đình Khoa (SE196737)
- Trịnh Phú Quốc (SE190287)
- Trần Đăng Khoa (SE194398)
- Đặng Đỗ Cao Sang (SE193269)
- Đào Lý Phi Hùng (SE172826)

Topic Code: RT-SWT-SE1905

Problem Statement: Manual Effort vs. Closed-Source Limitations

Manual Process

Creating Gherkin scenarios and Python Step Definitions is slow and error-prone.

 **High overhead; Human error**

State of the Art

Dependence on expensive closed-source APIs (GPT-4o); focus only on Gherkin generation.

 **High cost; Data privacy risks**

Research GAP (GAP-T)

No prior study systematically compares **Zero-Shot, Few-Shot, and CoT** for generating both Gherkin Scenarios and Python Step Definitions using local open-source LLMs.

 **Missing dual-output optimization**

CORE CHALLENGE



Existing automation lacks a **cost-effective, local workflow** that simultaneously generates both executable Gherkin scenarios and Python code.

Related Work (N = 37 Research Papers)

Paper	Tool/LLM	Dataset (Size)	Metric	Main Findings / Limitations	
Rathnayake 2026	GPT-4o (Zero-shot)	500 Real Stories	Cosine, BERTScore	High semantic similarity; fails to generate step definitions.	i
Fernandes 2025	GPT-4, Llama-3	34 Stories	METEOR	Only generates Gherkin; small single-domain sample.	i
Selfbehave 2026	LLaMA-3-8B (LoRA)	Synthetic Data	Syntax Accuracy	99.2% Gherkin accuracy; lacks real-world validity.	
Tesfalidet 2025	GPT-4	10-20 Stories	Executable Rate	85% executable steps; lacks dual static validation.	

! Research Gap & Conclusion

Current studies lack a comparative evaluation of Prompt Engineering (Zero-Shot, Few-Shot, CoT) for simultaneous Gherkin and Python generation using local open-source LLMs. Our research fills this gap by comparing these techniques while ensuring both executable syntax and semantic integrity.

Research Questions & Hypotheses

RQ1: Semantic Evaluate semantic quality of generated BDD artifacts.	RQ2: Syntax Assess Gherkin and Python Step Definition correctness.	RQ3: Prompting Compare Zero-Shot, Few-Shot, and CoT performance.
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Metric	Null Hypothesis (H_0)	Success Criteria	Statistical Test
Semantic (RQ1)	Similarity ≤ 0.80	≥ 0.80 Cosine Sim.	One-sample Wilcoxon
Syntax (RQ2)	$p(\text{syntax}) \leq 0.85$	$\geq 85\%$ Pass Rate	Binomial Exact Test
Prompting (RQ3)	No median difference	$p < 0.05$ Significance	Kruskal-Wallis

MODEL

Qwen2.5-7B-Instruct (Ollama)

TARGET

0.80 Similarity / 85% Syntax

SCOPE

100-sample sets per technique

Experimental Pipeline

A systematic dual-generation and validation workflow ensuring deterministic results for BDD automation.



MODEL CONFIG

Temperature = **0**, Top_p = **1**
(Deterministic Generation)

METRIC THRESHOLDS

Semantic Similarity $\geq$ **0.80**
Executable Syntax $\geq$ **85%**

STATISTICAL FRAMEWORK

Wilcoxon (Continuous Non-normal)
McNemar (Binary Rates)

Dataset & Model Configuration

Experimental Dataset

SOURCE

Multi-domain real dataset (Rathnayake et al. 2026)

TOTAL VOLUME

500 Stories

TEST SAMPLE

100 Stories

Sampling Strategy:

Simple Random Sample (Without replacement)

Random Seed: **42**

Model Configuration

CORE LLM

Qwen2.5-7B-Instruct

 Local Deployment via Ollama

TEMPERATURE

0

TOP_P

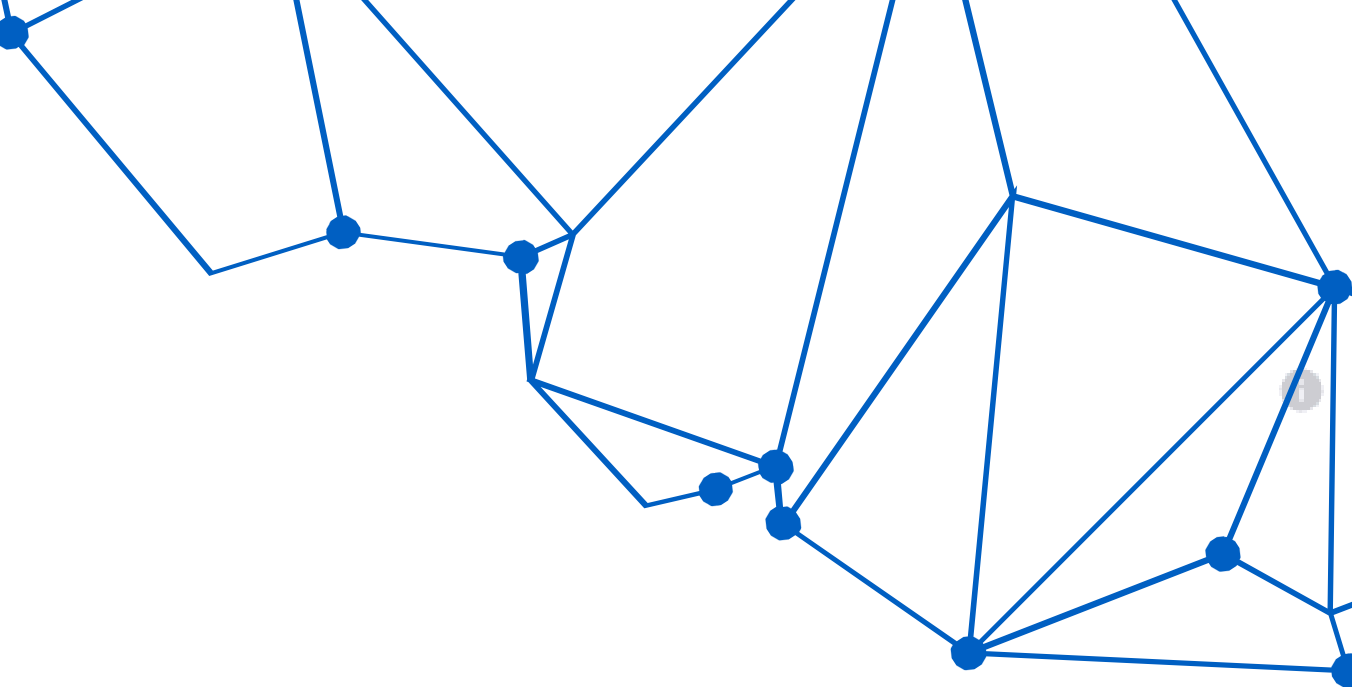
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Experimental Focus:

Ensuring [Deterministic Generation](#) and [Reproducibility](#) for BDD scenario and step definition evaluation across multiple business domains.



Statistical Plan: Evaluation Framework



Objective	Threshold	Statistical Test	Rationale
Semantic Similarity (RQ1)	≥ 0.80	One-sample Wilcoxon	Compare non-normal median against threshold for quality validation.
Executable Syntax (RQ2)	≥ 85%	Binomial Exact Test	Analyze binary PASS/FAIL proportion on a finite sample size (N=100).
Paired Semantic (RQ3)	Control	Paired Wilcoxon	Compare paired continuous non-normal data across prompt techniques.
Paired Syntax (RQ3)	Control	McNemar's Test	Assess dependent binary proportions for syntactic consistency differences.

Threats to Validity: Ensuring Scientific Rigor & Reliability

Internal Threat

POTENTIAL RISK

Bias in prompt design & Few-Shot example selection.

MITIGATION STRATEGY

Standardized templates; use of multi-domain BDD examples (Login, Cart).

External Threat

POTENTIAL RISK

Limited dataset coverage across diverse business domains.

MITIGATION STRATEGY

Utilization of a real-world dataset from 4 large enterprise projects.

Construct Threat

POTENTIAL RISK

SBERT cosine similarity vs. nuanced human judgment.

MITIGATION STRATEGY

Use of all-MiniLM-L6-v2; pre-filtering via dual static syntax validation.

Key Focus: To ensure findings are generalizable, we prioritize deterministic generation (Temp=0) and cross-sector data validation to minimize subjective LLM drift.

Timeline & Role Delegation

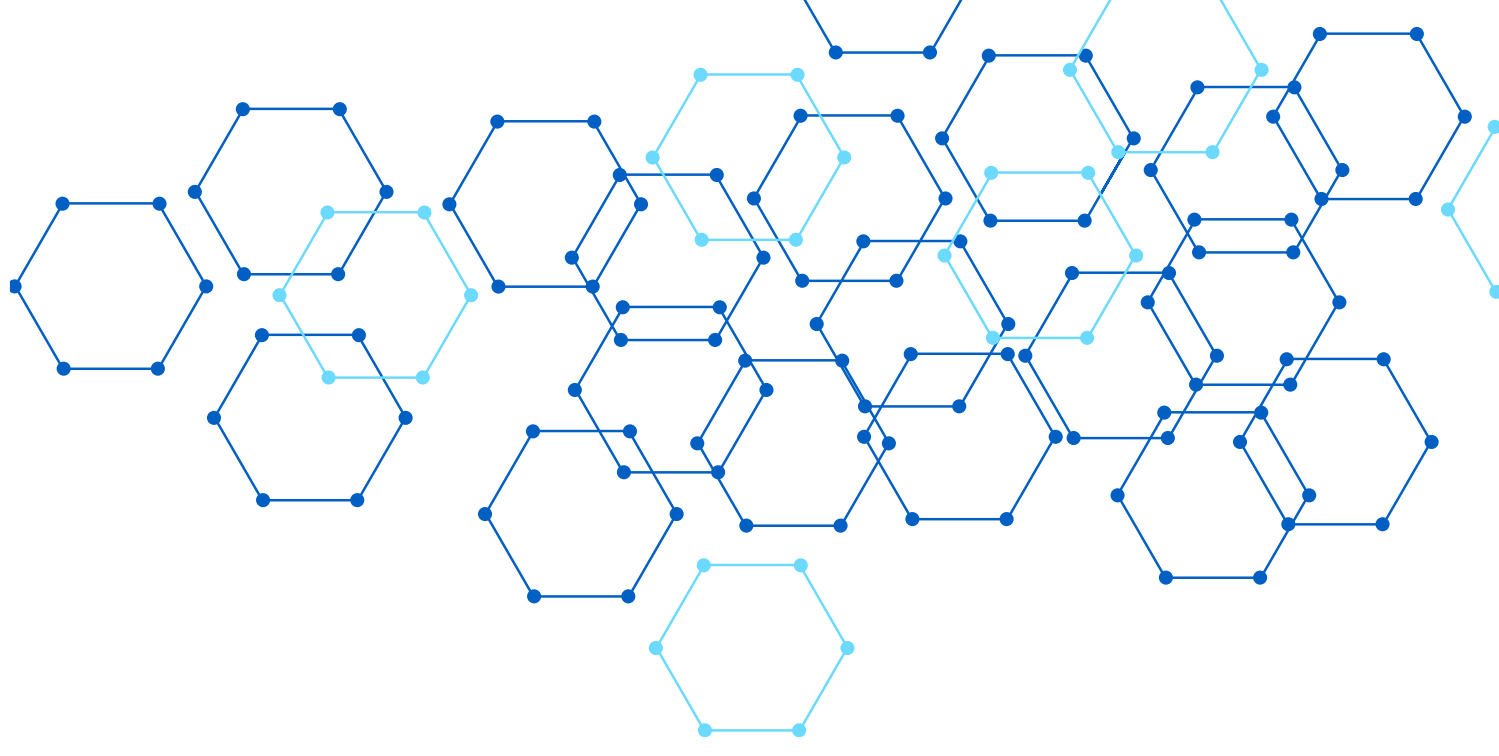
Team Responsibility Matrix

Role	Lead Member	Primary Responsibilities
Project Lead (PL)	Trần Đăng Khoa	Timeline coordination & document consistency review
Data & Ground Truth (DG)	Đào Lý Phi Hùng	100 sample extraction & ground truth verification
LLM Runner (LR)	 Trịnh Phú Quốc	Local Qwen deployment & prompt design execution
Metrics & Stats (MS)	Đặng Đỗ Cao Sang	Dual static parser programming & statistical testing
Report Writer (RW)	Ngô Đình Khoa	Threats analysis, visualization & defense preparation

PROJECT IMPLEMENTATION TIMELINE

June 2026





Expected Contributions

Research Novelty: Advancing Automated BDD Testing



Contribution 1 – Empirical Comparison

First systematic comparison of **Zero-Shot, Few-Shot, and Chain-of-Thought** prompting for dual BDD artifact generation using a local open-source LLM.

- Multi-prompting evaluation
- Dual artifact focus



Contribution 2 – Dual Validation

Evaluate generated outputs using both **semantic similarity** and **static syntax validation** (Gherkin Parser + Python AST).

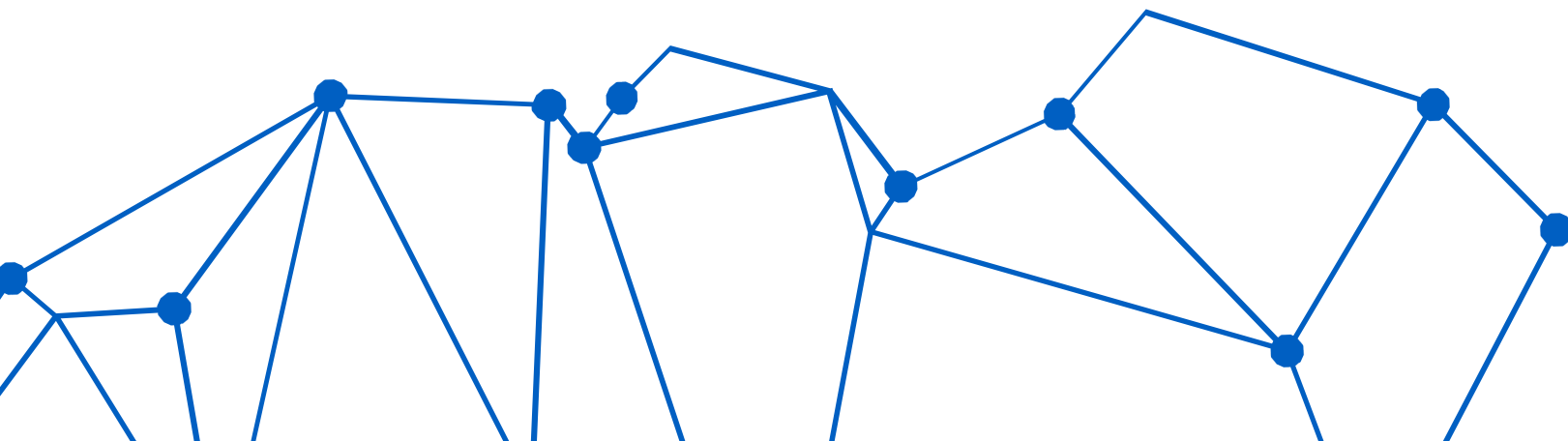
- Holistic quality assessment
- Structural & meaning alignment



Contribution 3 – Research Benchmark

Provide **reproducible experimental results** and benchmarking for local LLM-based automated BDD test generation.

- Open research methodology
- Local LLM performance baseline



Research GAP Mapping

EXISTING STUDIES	LIMITATIONS
Fernandes et al.	Only generates Gherkin scenarios
Rathnayake et al.	Does not generate Python Step Definitions
Kapurapu et al.	No semantic quality evaluation
Tesfalidet et al.	No dual static validation

💡 Addressing the Gaps (Research Novelty)

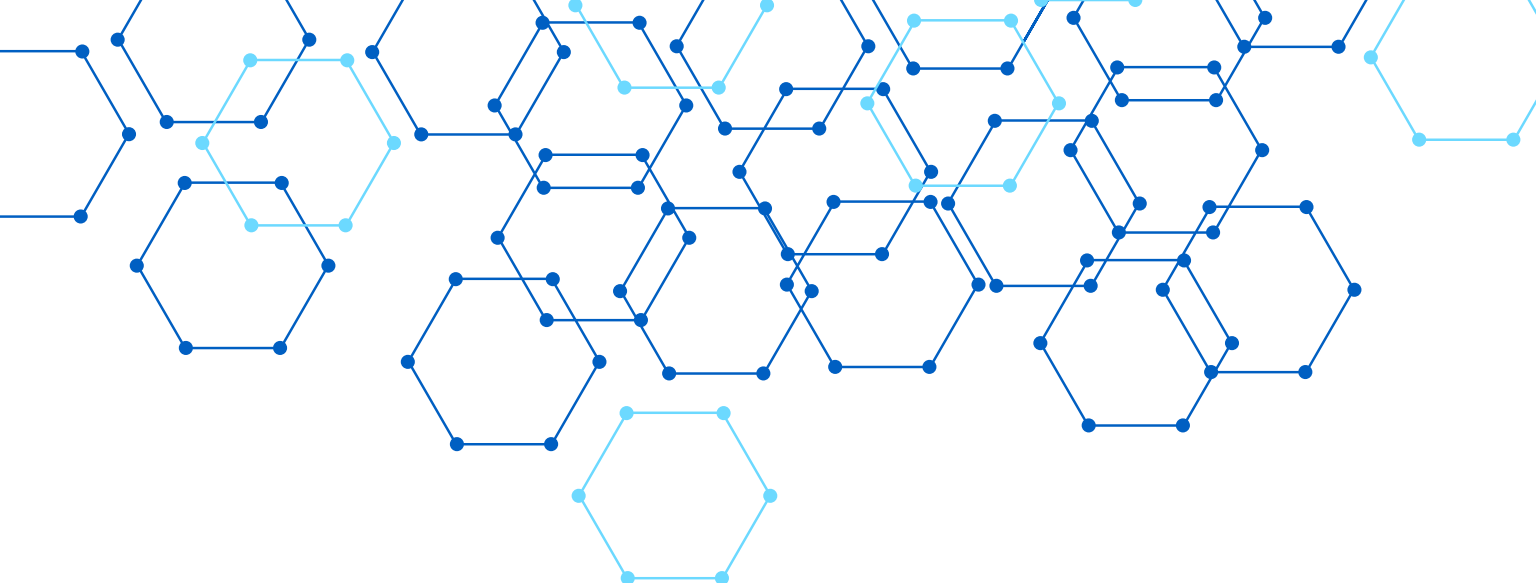
✓ Compare Prompt Engineering techniques (Zero-Shot, Few-Shot, CoT)	✓ Generate both Gherkin Scenarios and Python Step Definitions
✓ Apply dual validation (Semantic + Static Syntax)	✓ Evaluate using a local open-source LLM (Qwen2.5-7B)

Bảng minh chứng nghiên cứu liên quan

BACKUP

Related Work Evidence Table • Tổng hợp 8 công trình nghiên cứu liên quan trực tiếp đến sinh kịch bản Gherkin & Step Definitions

Công trình nghiên cứu	Công cụ / LLM	Tập dữ liệu (Quy mô)	Độ đo	Kết quả tốt nhất	Hạn chế
Rathnayake et al. 2026	GPT-4o (Zero-shot)	500 User Stories	BERTScore, Cosine Semantic Similarity	BERTScore = 91.16%, Cosine = 53.96%	Loại trừ sinh step definitions; không có kiểm định tĩnh kép.
Fernandes et al. 2025	GPT-3.5, GPT-4, LLaMA-3, Gemini, DeepSeek R1	34 User Stories	METEOR Score	METEOR = 0.84 (Gemini zero-shot)	Chỉ sinh kịch bản Gherkin; cỡ mẫu nhỏ, đơn miền.
Selfbehave et al. 2026	LLaMA-3-8B (LoRA-FT)	Tập dữ liệu tự sinh	Tỷ lệ cú pháp tĩnh	99.2% đúng cú pháp Gherkin	Hoàn toàn phụ thuộc vào dữ liệu tự sinh bằng AI.
Kapurapu et al. 2024	GPT-3.5, GPT-4, Llama-2	Cỡ mẫu nhỏ	Cú pháp qua gherkin-lint	100% đúng cú pháp (GPT-4)	Không sinh Step Definitions; không đánh giá ngữ nghĩa.
Tesfalidet et al. 2025	GPT-4	10–20 User Stories	Cosine Similarity, Behave run rate	Similarity = 0.81, Executable rate = 85%	Cỡ mẫu nhỏ; giới hạn miền FinTech; thiếu bộ kiểm định tĩnh kép.
dos Santos et al. 2026	ChatGPT, Gemini, Grok, Copilot	34 User Stories	Độ phủ tiêu chí nghiệm thu	Độ phủ cao cho ChatGPT	Quy mô dữ liệu nhỏ; chỉ sinh kịch bản Gherkin.
Mendoza et al. 2024	ChatGPT-4, Gemini, Copilot	5 User Stories	Thang đo Likert (đánh giá người)	Điểm chấp nhận người cao	Cỡ mẫu cực nhỏ (5 mẫu); đánh giá cảm quan.
Bergsmann et al. 2024	Multi-Agent LLMs	12–15 web apps	Tỷ lệ chạy test thành công	Tỷ lệ thành công cao	Dựa trên hệ thống multi-agent độ trễ cao, tốn token.



THANK YOU

Presented by Group 2

